

Integration Manual

for S32G LIN_LLCE Driver

Document Number: IM11LINLLCEASR4.4 Rev0000R1.0.7 Rev. 1.0

1 Revision History	2
2 Introduction	3
2.1 Supported Derivatives	3
2.2 Overview	4
2.3 About This Manual	4
2.4 Acronyms and definitions	5
2.5 Reference List	5
3 Building the driver	6
3.1 Build Options	6
3.1.1 GCC Compiler/Assembler/Linker Options	6
3.1.2 GHS Compiler/Assembler/Linker Options	9
3.2 Files required for compilation	11
3.3 Setting up the plugins	13
4 Function calls to module	15
4.1 Function calls during start-up	15
4.2 Function calls during shutdown	15
4.3 Function calls during wake-up	16
5 Module requirements	17
5.1 Exclusive areas to be defined in bsw scheduler	17
5.2 Unavailable exclusive areas	17
5.3 Peripheral hardware requirements	17
5.4 ISR to configure	17
5.5 ISR Macro	18
5.5.1 Without an Operating System	18
5.5.2 With an Operating System	18
5.6 Other autosar modules dependencies	19
5.7 Data cache restrictions	19
5.8 User Mode support	19
5.8.1 User Mode configuration in the module	19
5.8.2 User Mode configuration in AutosarOS	19
5.9 Multicore support.	20
6 Main API Requirements	21
6.1 Main function calls within BSW scheduler	21
6.2 API requirements	21
6.3 Call-back Notifications	21
7 Memory allocation	22



7.1 Sections to be defined in MemMap.h	22
7.2 Linker command file	23
8 Integration Steps	24
9 External assumptions for driver	25



Chapter 1

Revision History

Revision	Date	Author	Description
1.0	29.09.2023	NXP LLCE Team	Prepared for release S32G LLCE AUTOSAR 4.4 Version 1.0.7

Chapter 2

Introduction

- [Supported Derivatives](#)
- [Overview](#)
- [About This Manual](#)
- [Acronyms and definitions](#)
- [Reference List](#)

This integration manual describes the integration requirements for Lin_43_LLCE Driver for S32G microcontrollers.

2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of NXP Semiconductors:

- s32g274a_bga525
- s32g254a_bga525
- s32g233a_bga525
- s32g234m_bga525
- s32g378a_bga525
- s32g379a_bga525
- s32g398a_bga525
- s32g399a_bga525
- s32g338m_bga525
- s32g339m_bga525
- s32g358a_bga525
- s32g359a_bga525

All of the above microcontroller devices are collectively named as S32.

2.2 Overview

AUTOSAR (AUTomotive Open System ARchitecture) is an industry partnership working to establish standards for software interfaces and software modules for automobile electronic control systems.

AUTOSAR:

- paves the way for innovative electronic systems that further improve performance, safety and environmental friendliness.
- is a strong global partnership that creates one common standard: "Cooperate on standards, compete on implementation".
- is a key enabling technology to manage the growing electrics/electronics complexity. It aims to be prepared for the upcoming technologies and to improve cost-efficiency without making any compromise with respect to quality.
- facilitates the exchange and update of software and hardware over the service life of the vehicle.

2.3 About This Manual

This Technical Reference employs the following typographical conventions:

- **Boldface** style: Used for important terms, notes and warnings.
- *Italic* style: Used for code snippets in the text. Note that C language modifiers such "const" or "volatile" are sometimes omitted to improve readability of the presented code.

Notes and warnings are shown as below:

Note

This is a note.

Warning

This is a warning

2.4 Acronyms and definitions

Term	Definition
API	Application Programming Interface
ASM	Assembler
AUTOSAR	Automotive Open System Architecture
BSMI	Basic Software Make file Interface
C/CPP	C and C++ Source Code
DEM	Diagnostic Event Manager
DET	Default Error Tracer
EcuM	ECU state Manager
GUI	Graphical User Interface
ISR	Interrupt Service Routine
LIN	Local Interconnect Network
MCU	Micro Controller Unit
RTE	Run-Time Environment
N/A	Not Applicable
OS	Operating System
PB Variant	Post Build Variant
PC Variant	Pre Compile Variant
VLE	Variable Length Encoding

2.5 Reference List

#	Title	Version
1	Specification of LIN Driver	AUTOSAR Release 4.4.0
2	S32G2 Reference Manual	Rev. 5, 05/2022
3	S32G2 Data Sheet	Rev. 5, 05/2022
4	S32G2 Errata Document	S32G2 Mask Set Errata for Mask 0P77B, Rev. 3.1
5	S32G2 Errata Document	S32G2 Mask Set Errata for Mask 1P77B, Rev. 1.1
6	S32G3 Reference Manual	Rev. 2, 09/2022
7	S32G3 Data Sheet	Rev. 2, Draft C, 09/2021
8	S32G3 Errata Document	S32G3 Mask Set Errata for Mask 1P72B, Rev. 2.2

Chapter 3

Building the driver

- [Build Options](#)
- [Files required for compilation](#)
- [Setting up the plugins](#)

This section describes the source files and various compilers, linker options used for building the driver. It also explains the EB Tresos Studio plugin setup procedure.

3.1 Build Options

- [GCC Compiler/Assembler/Linker Options](#)
- [GHS Compiler/Assembler/Linker Options](#)

The RTD driver files are compiled using:

- NXP GCC 9.2.0 20190812 (Build 1649 Revision gaf57174) (for both S32G2 and S32G3)
- Green Hills Multi 7.1.6d / Compiler 2020.1.4 (for both S32G2 and S32G3)

The compiler, assembler, and linker flags used for building the driver are explained below.

The TS_T40D11M10I7R0 part of the plugin name is composed as follows:

- T = Target_Id (e.g. T40 identifies Cortex-M architecture)
- D = Derivative_Id (e.g. D11 identifies S32 platform)
- M = SW_Version_Major and SW_Version_Minor
- I = SW_Version_Patch
- R = Reserved

3.1.1 GCC Compiler/Assembler/Linker Options

3.1.1.1 GCC Compiler Options

Compiler Option	Description
-mcpu=cortex-m7	Targeted ARM processor for which GCC should tune the performance of the code
-mthumb	Generates code that executes in Thumb state
-mlittle-endian	Generate code for a processor running in little-endian mode
-mfpv=fpv5-sp-d16	Specifies the floating-point hardware available on the target
-mfloat-abi=hard	Specifies the floating-point ABI to use. "hard" allows generation of floating-point instructions and uses FPU-specific calling conventions
-std=c99	Specifies the ISO C99 base standard
-Os	Optimize for size. Enables all -O2 optimizations except those that often increase code size
-ggdb3	Produce debugging information for use by GDB using the most expressive format available, including GDB extensions if at all possible. Level 3 includes extra information, such as all the macro definitions present in the program
-Wall	Enables all the warnings about constructions that some users consider questionable, and that are easy to avoid (or modify to prevent the warning), even in conjunction with macros
-Wextra	This enables some extra warning flags that are not enabled by -Wall
-pedantic	Issue all the warnings demanded by strict ISO C. Reject all programs that use forbidden extensions. Follows the version of the ISO C standard specified by the aforementioned -std option
-Wstrict-prototypes	Warn if a function is declared or defined without specifying the argument types
-Wundef	Warn if an undefined identifier is evaluated in an #if directive. Such identifiers are replaced with zero
-Wunused	Warn whenever a function, variable, label, value, macro is unused
-Werror=implicit-function-declaration	Make the specified warning into an error. This option throws an error when a function is used before being declared
-Wsign-compare	Warn when a comparison between signed and unsigned values could produce an incorrect result when the signed value is converted to unsigned.
-Wdouble-promotion	Give a warning when a value of type float is implicitly promoted to double
-fno-short-enums	Specifies that the size of an enumeration type is at least 32 bits regardless of the size of the enumerator values.
-funsigned-char	Let the type char be unsigned by default, when the declaration does not use either signed or unsigned
-funsigned-bitfields	Let a bit-field be unsigned by default, when the declaration does not use either signed or unsigned
-fomit-frame-pointer	Omit the frame pointer in functions that don't need one. This avoids the instructions to save, set up and restore the frame pointer; on many targets it also makes an extra register available.

Compiler Option	Description
-fno-common	Makes the compiler place uninitialized global variables in the BSS section of the object file. This inhibits the merging of tentative definitions by the linker so you get a multiple-definition error if the same variable is accidentally defined in more than one compilation unit
-fstack-usage	This option is only used to build test for generation Ram/↔ Stack size report. Makes the compiler output stack usage information for the program, on a per-function basis
-fdump-ipa-all	This option is only used to build test for generation Ram/↔ Stack size report. Enables all inter-procedural analysis dumps
-c	Stop after assembly and produce an object file for each source file
-DS32XX	Predefine S32XX as a macro, with definition 1
-DS32G2XX	Predefine S32G2XX as a macro, with definition 1
-DS32G3XX	Predefine S32G3XX as a macro, with definition 1
-DS32R45	Predefine S32R45 as a macro, with definition 1
-DGCC	Predefine GCC as a macro, with definition 1
-DUSE_SW_VECTOR_MODE	Predefine USE_SW_VECTOR_MODE as a macro, with definition 1. By default, the drivers are compiled to handle interrupts in Software Vector Mode
-DD_CACHE_ENABLE	Predefine D_CACHE_ENABLE as a macro, with definition 1. Enables data cache initialization in source file system.↔ c under the Platform driver
-DI_CACHE_ENABLE	Predefine I_CACHE_ENABLE as a macro, with definition 1. Enables instruction cache initialization in source file system.c under the Platform driver
-DENABLE_FPU	Predefine ENABLE_FPU as a macro, with definition 1. Enables FPU initialization in source file system.c under the Platform driver
-DMCAL_ENABLE_USER_MODE_SUPPORT	Predefine MCAL_ENABLE_USER_MODE_SUPPORT↔ RT as a macro, with definition 1. Allows drivers to be configured in user mode.

3.1.1.2 GCC Assembler Options

Assembler Option	Description
-X assembler-with-cpp	Specifies the language for the following input files (rather than letting the compiler choose a default based on the file name suffix)
-mcpu=cortexm7	Targeted ARM processor for which GCC should tune the performance of the code
-mfpu=fpv5-sp-d16	Specifies the floating-point hardware available on the target
-mfloat-abi=hard	Specifies the floating-point ABI to use. "hard" allows generation of floating-point instructions and uses FPU-specific calling conventions
-mthumb	Generates code that executes in Thumb state
-c	Stop after assembly and produce an object file for each source file

3.1.1.3 GCC Linker Options

Linker Option	Description
-Wl,-Map,filename	Produces a map file
-T linkerfile	Use linkerfile as the linker script. This script replaces the default linker script (rather than adding to it)
-entry=Reset_Handler	Specifies that the program entry point is Reset_Handler
-nostartfiles	Do not use the standard system startup files when linking
-mcpu=cortexm7	Targeted ARM processor for which GCC should tune the performance of the code
-mthumb	Generates code that executes in Thumb state
-mfpu=fpv5-sp-d16	Specifies the floating-point hardware available on the target
-mfloat-abi=hard	Specifies the floating-point ABI to use. "hard" allows generation of floating-point instructions and uses FPU-specific calling conventions
-mlittle-endian	Generate code for a processor running in little-endian mode
-ggdb3	Produce debugging information for use by GDB using the most expressive format available, including GDB extensions if at all possible. Level 3 includes extra information, such as all the macro definitions present in the program
-lc	Link with the C library
-lm	Link with the Math library
-lgcc	Link with the GCC library

3.1.2 GHS Compiler/Assembler/Linker Options

3.1.2.1 GHS Compiler Options

Compiler Option	Description
-cpu=cortexm7	Selects target processor: Arm Cortex M7
-thumb	Selects generating code that executes in Thumb state
-fpu=vfpv5_d16	Specifies hardware floating-point using the v5 version of the VFP instruction set, with 16 double-precision floating-point registers
-fsingle	Use hardware single-precision, software double-precision FP instructions
-C99	Use (strict ISO) C99 standard (without extensions)
-ghstd=last	Use the most recent version of Green Hills Standard mode (which enables warnings and errors that enforce a stricter coding standard than regular C and C++)
-Osize	Optimize for size
-gnu_asm	Enables GNU extended asm syntax support
-dual_debug	Generate DWARF 2.0 debug information
-G	Generate debug information
-keeptempfiles	Prevents the deletion of temporary files after they are used. If an assembly language file is created by the compiler, this option will place it in the current directory instead of the temporary directory

Compiler Option	Description
-Wimplicit-int	Produce warnings if functions are assumed to return int
-Wshadow	Produce warnings if variables are shadowed
-Wtrigraphs	Produce warnings if trigraphs are detected
-Wundef	Produce a warning if undefined identifiers are used in #if preprocessor statements
-unsigned_chars	Let the type char be unsigned, like unsigned char
-unsigned_fields	Bitfields declared with an integer type are unsigned
-no_commons	Allocates uninitialized global variables to a section and initializes them to zero at program startup
-no_exceptions	Disables C++ support for exception handling
-no_slash_comment	C++ style // comments are not accepted and generate errors
-prototype_errors	Controls the treatment of functions referenced or called when no prototype has been provided
-incorrect_pragma_warnings	Controls the treatment of valid #pragma directives that use the wrong syntax
-c	Stop after assembly and produce an object file for each source file
-DS32XX	Predefine S32XX as a macro, with definition 1
-DS32G2XX	Predefine S32G2XX as a macro, with definition 1
-DS32G3XX	Predefine S32G3XX as a macro, with definition 1
-DS32R45	Predefine S32R45 as a macro, with definition 1
-DGHS	Predefine GHS as a macro, with definition 1
-DUSE_SW_VECTOR_MODE	Predefine USE_SW_VECTOR_MODE as a macro, with definition 1. By default, the drivers are compiled to handle interrupts in Software Vector Mode
-DD_CACHE_ENABLE	Predefine D_CACHE_ENABLE as a macro, with definition 1. Enables data cache initialization in source file system.↵c under the Platform driver
-DI_CACHE_ENABLE	Predefine I_CACHE_ENABLE as a macro, with definition 1. Enables instruction cache initialization in source file system.c under the Platform driver
-DENABLE_FPU	Predefine ENABLE_FPU as a macro, with definition 1. Enables FPU initialization in source file system.c under the Platform driver
-DMCAL_ENABLE_USER_MODE_SUPPORT	Predefine MCAL_ENABLE_USER_MODE_SUPPORT↵RT as a macro, with definition 1. Allows drivers to be configured in user mode

3.1.2.2 GHS Assembler Options

Assembler Option	Description
-cpu=cortexm7	Selects target processor: Arm Cortex M7
-fpu=vfpv5_d16	Specifies hardware floating-point using the v5 version of the VFP instruction set, with 16 double-precision floating-point registers
-fsingle	Use hardware single-precision, software double-precision FP instructions
-preprocess_assembly_files	Controls whether assembly files with standard extensions such as .s and .asm are preprocessed

Assembler Option	Description
-list	Creates a listing by using the name and directory of the object file with the .lst extension
-c	Stop after assembly and produce an object file for each source file

3.1.2.3 GHS Linker Options

Linker Option	Description
-e Reset_Handler	Make the symbol Reset_Handler be treated as a root symbol and the start label of the application
-T linker_script_file.ld	Use linker_script_file.ld as the linker script. This script replaces the default linker script (rather than adding to it)
-map	Produce a map file
-keepmap	Controls the retention of the map file in the event of a link error
-Mn	Generates a listing of symbols sorted alphabetically/numerically by address
-delete	Instructs the linker to remove functions that are not referenced in the final executable. The linker iterates to find functions that do not have relocations pointing to them and eliminates them
-ignore_debug_references	Ignores relocations from DWARF debug sections when using -delete. DWARF debug information will contain references to deleted functions that may break some third-party debuggers
-Llibrary_path	Points to library_path (the libraries location) for thumb2 to be used for linking
-larch	Link architecture specific library
-lstartup	Link run-time environment startup routines. The source code for the modules in this library is provided in the src/libstartup directory
-lind_sd	Link language-independent library, containing support routines for features such as software floating point, run-time error checking, C99 complex numbers, and some general purpose routines of the ANSI C library
-v	Prints verbose information about the activities of the linker, including the libraries it searches to resolve undefined symbols
-nostartfiles	Controls the start files to be linked into the executable

3.2 Files required for compilation

This section describes the include files required to compile, assemble (if assembler code) and link the LIN_43_LLCE driver for microcontrollers.

To avoid integration of incompatible files, all the include files from other modules shall have the same AR_MAJOR_VERSION and AR_MINOR_VERSION, i.e. only files with the same AUTOSAR major and minor versions can be compiled.

- LIN_43_LLCE Files
 - ../Lin_43_LLCE_TS_T40D11M10I6R0/src/Lin_43_LLCE.c
 - ../Lin_43_LLCE_TS_T40D11M10I6R0/src/Lin_Llce.c

- `../Lin_43_LLCE_TS_T40D11M10I6R0/src/Lin_Llce_Irq.c`
- `../Lin_43_LLCE_TS_T40D11M10I6R0/include/Lin_43_LLCE.h`
- `../Lin_43_LLCE_TS_T40D11M10I6R0/include/Lin_43_LLCE_IPW.h`
- `../Lin_43_LLCE_TS_T40D11M10I6R0/include/Lin_Llce.h`
- `../Lin_43_LLCE_TS_T40D11M10I6R0/include/Llce_InterfaceCore2Core.h`
- `../Lin_43_LLCE_TS_T40D11M10I6R0/include/Llce_InterfaceLinTypes.h`

- LIN_43_LLCE Generated Files

- `Lin_43_LLCE_[VariantName]_PBcfg.c` - This file should be generated by the user using a configuration tool for compilation. The file contains the definition of the init pointer for the respective variant.
- `Lin_43_LLCE_Cfg.c` - This file should be generated by the user using a configuration tool for compilation.
- `Lin_43_LLCE_Cfg.h` - This file should be generated by the user using a configuration tool for compilation.

NOTE As a deviation from standard

- `Lin_43_LLCE_[VariantName]_PBcfg.c` files will contain the definition for all parameters that are variant aware, independent of the configuration class that will be selected (PC, PB)
- `Lin_43_LLCE_Cfg.c` file will contain the definition for all configuration structures containing only variables that are not variant aware, configured and generated only once. This file alone does not contain the whole structure needed by `Lin_Init` function to configure the driver. Based on the number of variants configured in the `EcuC`, there can be more than one configuration structure for one module even for PreCompile variant.
- Files from BaseNXP common folder
 - `../BaseNXP_TS_T40D11M10I6R0/generate_PC/include/modules.h`
 - `../BaseNXP_TS_T40D11M10I6R0/generate_PC/include/Reg_eSys.h`
 - `../BaseNXP_TS_T40D11M10I6R0/include/Compiler.h`
 - `../BaseNXP_TS_T40D11M10I6R0/include/CompilerDefinition.h`
 - `../BaseNXP_TS_T40D11M10I6R0/include/Compiler_Cfg.h`
 - `../BaseNXP_TS_T40D11M10I6R0/include/ComStack_Cfg.h`
 - `../BaseNXP_TS_T40D11M10I6R0/include/ComStackTypes.h`
 - `../BaseNXP_TS_T40D11M10I6R0/include/Lin_GeneralTypes.h`
 - `../BaseNXP_TS_T40D11M10I6R0/include/Mcal.h`
 - `../BaseNXP_TS_T40D11M10I6R0/include/Lin_43_LLCE_MemMap.h`
 - `../BaseNXP_TS_T40D11M10I6R0/include/Platform_Types.h`
 - `../BaseNXP_TS_T40D11M10I6R0/include/Reg_Macros.h`
 - `../BaseNXP_TS_T40D11M10I6R0/include/Std_Types.h`
 - `../BaseNXP_TS_T40D11M10I6R0/include/StandardTypes.h`
 - `../BaseNXP_TS_T40D11M10I6R0/include/Soc_Ips.h`
- Files from Dem folder:
 - `../Dem_TS_T40D11M10I6R0/generate_PC/include/Dem_IntErrId.h`
 - `../Dem_TS_T40D11M10I6R0/include/Dem.h`
 - `../Dem_TS_T40D11M10I6R0/include/Dem_Types.h`

- Files from Det folder:
 - ../Det_TS_T40D11M10I6R0/include/Det.h
- Files from EcuM folder:
 - ../EcuM_TS_T40D11M10I6R0/generate_PC/include/EcuM_Cfg.h
 - ../EcuM_TS_T40D11M10I6R0/include/EcuM.h
 - ../EcuM_TS_T40D11M10I6R0/include/EcuM_Externals.h
- Files from LinIf folder:
 - ../LinIfTS_T40D11M10I6R0/include/LinIf.h
 - ../LinIfTS_T40D11M10I6R0/include/LinIf_Cbk.h

3.3 Setting up the plugins

The LIN_43_LLCE driver was designed to be configured by using the EB Tresos Studio (version 27.0.1 or later.)

- Location of various files inside the LIN module folder:
 - VSMD (Vendor Specific Module Definition) file in EB tresos Studio XDM format:
 - * ../BaseNXP_TS_T40D11M10I6R0/config/BaseNXP.xdm
 - * ../EcuM_TS_T40D11M10I6R0/config/EcuM.xdm
 - * ../EcuC_TS_T40D11M10I6R0/config/EcuC.xdm
 - * ../Lin_43_LLCE_TS_T40D11M10I4R0/config/Lin_43_LLCE.xdm
 - * ../Mcu_TS_T40D11M10I6R0/config/Mcu.xdm
 - * ../Resource_TS_T40D11M10I6R0/config/Resource.xdm
 - VSMD (Vendor Specific Module Definition) file(s) in AUTOSAR compliant EPD format:
 - * ../BaseNXP_TS_T40D11M10I6R0/autosar/BaseNXP.epd
 - * ../EcuM_TS_T40D11M10I6R0/autosar/EcuM.epd
 - * ../EcuC_TS_T40D11M10I6R0/autosar/EcuC.epd
 - * ../Lin_43_LLCE_TS_T40D11M10I4R0/autosar/Lin_43_LLCE.epd
 - * ../Mcu_TS_T40D11M10I6R0/autosar/Mcu.epd
 - * ../Resource_TS_T40D11M10I6R0/autosar/Resource.epd
 - Code Generation Templates for parameters without variation points:
 - * ../Lin_43_LLCE_TS_T40D11M10I4R0/generate_PC/src/Lin_43_LLCE_Cfg.c
 - * ../Lin_43_LLCE_TS_T40D11M10I4R0/generate_PC/include/Lin_43_LLCE_Cfg.h
 - * ../Lin_43_LLCE_TS_T40D11M10I4R0/generate_PC/Lin_43_LLCE_VersionCheck_Inc.m
 - * ../Lin_43_LLCE_TS_T40D11M10I4R0/generate_PC/Lin_43_LLCE_VersionCheck_Src.m
 - * ../Lin_43_LLCE_TS_T40D11M10I4R0/generate_PC/Lin_43_LLCE_BaudRate_Comp.m
 - Code Generation Templates for variant aware parameters:
 - * ../Lin_43_LLCE_TS_T40D11M10I4R0/generate_PB/src/Lin_43_LLCE_PBcfg.c
 - * ../Lin_43_LLCE_TS_T40D11M10I4R0/generate_PB/Lin_43_LLCE_VersionCheck_Inc.m
 - * ../Lin_43_LLCE_TS_T40D11M10I4R0/generate_PB/Lin_43_LLCE_VersionCheck_Src_PB.m
 - * ../Lin_43_LLCE_TS_T40D11M10I4R0/generate_PB/Lin_43_LLCE_BaudRate_Comp.m
- Steps to generate the configuration:

1. Copy the module folders Lin_43_LLCE_TS_T40D11M10I4R0, BaseNXP_TS_T40D11M10I6R0, Ecu↔M_TS_T40D11M10I6R0, EcuC_TS_T40D11M10I6R0, Rte_TS_T40D11M10I6R0, Mcu_ TS_T40↔D11M10I6R0 into the Tresos plugins folder.
 2. Set the desired Tresos Output location folder for the generated sources and header files.
 3. Use the EB tresos Studio GUI to modify ECU configuration parameters values.
 4. Generate the configuration files.
- Dependencies
 - **MCU** is required to use System Clock when clock source is used as Peripheral clock source to generate LIN Segment values.
 - **ECUM** is required for selecting the reference to the wakeup source for every Lin controller.
 - **ECUC** is required for selecting the postbuild variant criterion.
 - **DET** is required for signaling the default error detection (parameters out of range, null pointers, etc).
 - **DEM** is required for signaling the production error detection (hardware failure, etc).
 - **RTE** is required for implementing data consistency of exclusive areas.

Chapter 4

Function calls to module

- Function calls during start-up
- Function calls during shutdown
- Function calls during wake-up

4.1 Function calls during start-up

LIN_43_LLCE shall be initialized during STARTUP phase of EcuM initialization. The API to be called for this is LIN_43_LLCE_Init(). The MCU module should be initialized before the LIN_43_LLCE is initialized. The Lin driver does not need OS Support except for ISR's. Hence, can be initialized either in STARTUP1 or STARTUP2 phase of EcuM initialization. This depends on the implementation, desired duration for STARTUP1 & Target hardware design. The LIN module shall be initialized by LIN_43_LLCE_Init(<&Lin_Configuration>) service call during the start-up before the LIN peripherals are used. Please note that GPIO pins used for connection of LIN physical layer have to be properly assigned to desired LINFlex module prior the LIN initialization: so the MCU and PORT modules shall be initialized before LIN is initialized. After the LIN module is initialized each LIN channel have to be initialized as well before using it. This is also done by the LIN_43_LLCE_Init(<&Lin_Configuration>) service.

4.2 Function calls during shutdown

There is no shutdown specific procedure for Lin driver. LIN driver can go to sleep mode using the following API services:

Lin_43_LLCE_GoToSleepInternal(LIN_CHANNEL): which put the LIN driver into sleep mode without sending of Go-to-sleep command over the bus.

Lin_43_LLCE_GoToSleep(LIN_CHANNEL): which put the LIN driver into sleep mode and send a Go-to-sleep command over the bus.

4.3 Function calls during wake-up

LIN driver supports the transmission of wake up command via the LIN bus. For this purposes the `Lin_43_LLCE_Wakeup(LIN_CHANNEL)` API service may be used.

External Wakeup**:

The Lin driver supports external wake-up from the bus in 2 modes:

- Upon wakeup detection on Rx pin of the configured LINFlex channel the ISR "Lin_CmdFromLlceIsr" will be executed (based on the LIN Channel configured) to wake-up Lin Driver.
- The Lin stack may call `Lin_43_LLCE_CheckWakeup(LIN_CHANNEL)` service API in order to identify if a slave on the Lin bus issued a wake-up request.

Chapter 5

Module requirements

- Exclusive areas to be defined in bsw scheduler
- Unavailable exclusive areas
- Peripheral hardware requirements
- ISR to configure
- ISR Macro
- Other autosar modules dependencies
- Data cache restrictions
- User Mode support
- Multicore support.

5.1 Exclusive areas to be defined in bsw scheduler

None.

5.2 Unavailable exclusive areas

None.

5.3 Peripheral hardware requirements

The LIN physical interface should be connected to the LIN module pins in order to get the LIN bus voltage levels.

5.4 ISR to configure

The following ISR's are used by the LIN driver:

Module requirements

ISR Name	NVIC interrupt vector	GIC interrupt vector
Host Interrupt from Core1	171	?

1. The type number and interrupt vectors of hardware channels are specific for each platform. See the documentation for details.
2. In case of AUTOSAR_OS_NOT_USED and the user wants to use the INTC in software mode, the compiler option "-DUSE_SW_VECTOR_MODE" has to be added to the list of compiler options.
3. NO external wake-up is supported by drivers. To wake-up the core from Standby 0 over LIN channels, the ICU drivers have to be used.

5.5 ISR Macro

RTD drivers use the ISR macro to define the functions that will process hardware interrupts. Depending on whether the OS is used or not, this macro can have different definitions.

5.5.1 Without an Operating System The macro *USING_OS_AUTOSAROS* must not be defined.

5.5.1.1 Using Software Vector Mode

The macro *USE_SW_VECTOR_MODE* must be defined and the ISR macro is defined as:

```
#define ISR(IsrName) void IsrName(void)
```

In this case, the drivers' interrupt handlers are normal C functions and their prologue/epilogue will handle the context save and restore.

5.5.1.2 Using Hardware Vector Mode

The macro *USE_SW_VECTOR_MODE* must not be defined and the ISR macro is defined as:

```
#define ISR(IsrName) INTERRUPT_FUNC void IsrName(void)
```

In this case, the drivers' interrupt handlers must also handle the context save and restore.

5.5.2 With an Operating System Please refer to your OS documentation for description of the ISR macro.

5.6 Other autosar modules dependencies

BaseNXP:** The BASENXP module contains the common files/definitions needed by all MCAL modules. Mcu:** The MCU driver provides services for basic microcontroller initialization, power down functionality, reset and microcontroller specific functions required by other MCAL software modules. The clocks need to be initialized prior to using the LIN driver. This module is required for setting the "LIN Clock Reference" value. Port:** The PORT module is used to configure the port pins with the needed modes, before they are used by the LIN module. For each LINFlex, the TXD and RXD signals need to be configured. EcuC:** The ECUC module is used for ECU configuration. MCAL modules need ECUC to retrieve the variant information. Det:** The DET module is used for enabling Default error detection. The API function used is Det_ReportError(). The activation/deactivation of Default error detection is configurable using the 'LinDevErrorDetect' configuration parameter. Dem:** The DEM module is used for enabling reporting of production relevant error status. The API function used is Dem_ReportErrorStatus(). EcuM:** This module is used for processing the Wakeup notifications of LIN. Whenever the module is in 'Sleep' mode and a wakeup event occurs on a wakeup capable channel, it is reported to EcuM through the EcuM_CheckWakeupEvent() API. This is configurable using the 'LinChannelWakeupInfo' configuration parameter. RTE:** The RTE module is needed for implementing data consistency of exclusive areas that are used by LIN module.

5.7 Data cache restrictions

None.

5.8 User Mode support

- [User Mode configuration in the module](#)
- [User Mode configuration in AutosarOS](#)

Lin module does not include registers protection. So, it is accessible to all registered in any public mode.

5.8.1 User Mode configuration in the module The LIN_43_LLCE driver can be run in user mode or supervisor mode without any config. User mode is not needed for this driver.

5.8.2 User Mode configuration in AutosarOS

When User mode is enabled, the driver may has the functions that need to be called as trusted functions in AutosarOS context. Those functions are already defined in driver and declared in the header <IpName>_IpTrustedFunctions.h. This header also included all headers files that contains all types definition used by parameters or return types of those functions. Refer the chapter [User Mode configuration in the module](#) for more detail about those functions and the name of header files they are declared inside. Those functions will be called indirectly with the naming convention below in order to AutosarOS can call them as trusted functions.

```
Call_<Function_Name>_TRUSTED(parameter1,parameter2,...)
```

That is the result of macro expansion `OsIf_Trusted_Call` in driver code:

```
#define OsIf_Trusted_Call[1-6params](name,param1,...,param6) Call_##name##_TRUSTED(param1,...,param6)
```

So, the following steps need to be done in AutosarOS:

Module requirements

- Ensure `MCAL_ENABLE_USER_MODE_SUPPORT` macro is defined in the build system or somewhere global.
- Define and declare all functions that need to call as trusted functions follow the naming convention above in Integration/User code. They need to be visible in `Os.h` for the driver to call them. They will do the marshalling of the parameters and call `CallTrustedFunction()` in OS specific manner.
- `CallTrustedFunction()` will switch to privileged mode and call `TRUSTED_<Function_Name>()`.
- `TRUSTED_<Function_Name>()` function is also defined and declared in Integration/User code. It will un-marshalling of the parameters to call `<Function_Name>()` of driver. The `<Function_Name>()` functions are already defined in driver and declared in `<IpName>_Ip_TrustedFunctions.h`. This header should be included in OS for OS call and indexing these functions.

See the sequence chart below for an example calling `Linflexd_Uart_Ip_Init_Privileged()` as a trusted function.

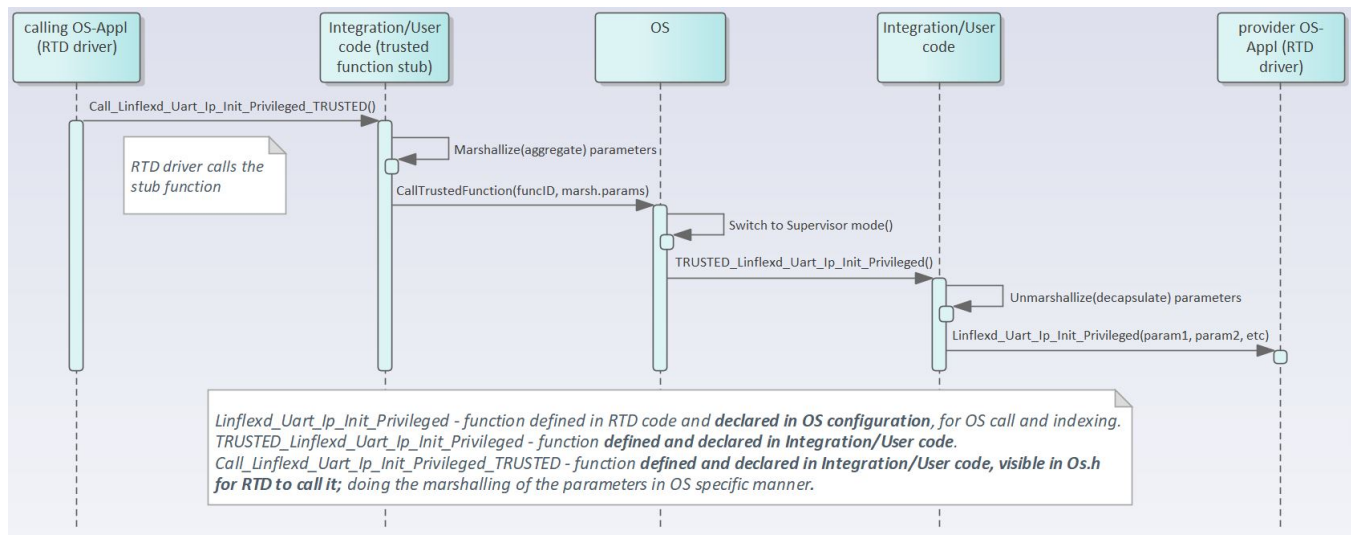


Figure 5.1 Example sequence chart for calling `Linflexd_Uart_Ip_Init_Privileged` as trusted function

5.9 Multicore support.

`Lin_43_LLCE` driver can only run on a single core.

Chapter 6

Main API Requirements

- [Main function calls within BSW scheduler](#)
- [API requirements](#)
- [Call-back Notifications](#)

6.1 Main function calls within BSW scheduler

None.

6.2 API requirements

Not Applicable.

6.3 Call-back Notifications

The LIN Driver uses 3 callback functions which have to be provided by the respective module: EcuM_Validate↔WakeupEvent(), EcuM_SetWakeupEvent() and EcuM_CheckWakeup() have to be provided by the EcuM module.

User Notification** None

Chapter 7

Memory allocation

- [Sections to be defined in MemMap.h](#)
- [Linker command file](#)

7.1 Sections to be defined in MemMap.h

Section name	Type of section	Description
LIN_43_LLCE_START_SEC_CONFIG_DATA_UNSPECIFIED↔	Configuration Data	Start of Memory Section for Config Data
LIN_43_LLCE_STOP_SEC_CONFIG_DATA_UNSPECIFIED↔	Configuration Data	End of Memory Section for Config Data
LIN_43_LLCE_START_SEC_CODE	Code	Start of memory Section for Code
LIN_43_LLCE_STOP_SEC_CODE	Code	End of memory Section for Code
LIN_43_LLCE_START_SEC_VARIABLES_CLEARED_8↔	Variables	Used for variables which have to be aligned to 8 bit. For instance used for variables of size 8 bit or used for composite data types: arrays, structs containing elements of maximum 8 bits. These variables are never cleared and never initialized by start-up code.
LIN_43_LLCE_STOP_SEC_VARIABLES_CLEARED_8↔	Variables	End of above section.
LIN_43_LLCE_START_SEC_VARIABLES_CLEARED_UNSPECIFIED↔	Variables	Used for variables, structures, arrays when the SIZE (alignment) does not fit the criteria of 8,16 or 32 bit. These variables are never cleared and never initialized by start-up code.
LIN_43_LLCE_STOP_SEC_VARIABLES_CLEARED_UNSPECIFIED↔	Variables	End of above section.
LIN_43_LLCE_START_SEC_VARIABLES_INIT_8↔	Variables	Used for variables which have to be aligned to 8 bit. For instance used for variables of size 8 bit or used for composite data types: arrays, structs containing elements of maximum 8 bits. These variables are initialized with values after every reset.

Section name	Type of section	Description
LIN_43_LLCE_STOP_SEC_VAR_IT_8	Variables	End of above section.
LIN_43_LLCE_START_SEC_VAR_IT_UNSPECIFIED	Variables	Used for variables, structures, arrays, when the SIZE (alignment) does not fit the criteria of 8,16 or 32 bit. These variables are initialized with values after every reset.
LIN_43_LLCE_STOP_SEC_VAR_IT_UNSPECIFIED	Variables	End of above section.
LIN_43_LLCE_START_SEC_CONST_32	Constant Data	Used for constants that have to be aligned to 32 bit.
LIN_43_LLCE_STOP_SEC_CONST_32	Constant Data	End of above section.
LIN_43_LLCE_START_SEC_CONST_UNSPECIFIED	Constant Data	Used for constants when the SIZE (alignment) does not fit the criteria of 8,16 or 32 bit.
LIN_43_LLCE_STOP_SEC_CONST_UNSPECIFIED	Constant Data	End of above section.
LIN_43_LLCE_START_SEC_SHAREDMEMORY	Variables	Used for variables, structures, arrays stored into shared memory.
LIN_43_LLCE_STOP_SEC_SHAREDMEMORY	Variables	End of above section.

7.2 Linker command file

Memory shall be allocated for every section defined in the driver's "<Module>_MemMap.h.

In addition, some sections are intended to be placed in Shared RAM, for communication with LLCE Firmware. These sections shall be placed in the linker command file in exactly the manner specified:

Section in linker file	Start address	Size (bytes)
LLCE_LIN_SHAREDMEMORY	0x4383D000	0x3000

The host application shall not access at all shared memory area before to be configured(set clock tree, etc.). For example, the loader needs to know that the shared memory of the LLCE must not be loaded. This can be done by specifying '(NOLOAD)' directive when using GCC compiler. Some compilers might do this implicitly (e.g GreenHills), but others don't (e.g GCC).

Chapter 8

Integration Steps

This section gives a brief overview of the steps needed for integrating this module:

1. Generate the required module configuration(s). For more details refer to section [Files Required for Compilation](#)
2. Allocate the proper memory sections in the driver's memory map header file ("`<Module>_MemMap.h`") and linker command file. For more details refer to section [Sections to be defined in `<Module>\_MemMap.h`](#)
3. Compile & build the module with all the dependent modules. For more details refer to section [Building the Driver](#)

Chapter 9

External assumptions for driver

The section presents requirements that must be complied with when integrating the LIN_LLCE driver into the application.

External Assumption Req ID	External Assumption Text
SWS_Lin_00045	One LIN driver provides access to one LIN hardware unit type (simple U↔ART or dedicated LIN hardware) that may consist of several LIN channels.
SWS_Lin_00225	There must be at least one statically defined configuration set available for the LIN driver. When the EcuM invokes the initialization function, it has to provide a specific pointer to the configuration that it wishes to use.
SWS_Lin_00014	Each LIN PID shall be associated with a checksum model (either 'enhanced' where the PID is included in the checksum, or 'classic' where only the response data is check-summed) (see Lin_PduType).
SWS_Lin_00015	Each LIN PID shall be associated with a response data length in bytes (see Lin_PduType).
SWS_Lin_00210	For LIN Master nodes, the upper layer of the LIN Driver has to keep the buffer data consistent until return of function call. For LIN Slave nodes, the upper layer of the LIN Driver has to keep the buffer data consistent until end of response transmission.
SWS_Lin_00246	If different LIN drivers are used, only one instance of this file has to be included in the source tree. For implementation all Lin_GeneralTypes.h related types in the documents mentioned before shall be considered.
SWS_Lin_00228	Name: - Lin_FramePidType - Type: - uint8 - Range: - 0...0xFE - - - The LIN identifier (0...0x3F) together with its two parity bits. - Description↔: - Represents all valid protected identifier used by Lin_SendFrame(). - Available via: - Lin_GeneralTypes.h -
SWS_Lin_00229	Name: - Lin_FrameCsModelType - Type: - Enumeration - Range: - LIN↔_ENHANCED_CS - - - Enhanced checksum model - LIN_CLASSIC_CS - - - Classic checksum model - Description: - This type is used to specify the Checksum model to be used for the LIN Frame. - Available via: - Lin↔_GeneralTypes.h -
SWS_Lin_00230	Name: - Lin_FrameResponseType - Type: - Enumeration - Range: - L↔IN_FRAMERESPONSE_TX - - - Response is generated from this node - LIN_FRAMERESPONSE_RX - - - Response is generated from another node and is relevant for this node. - LIN_FRAMERESPONSE_IGNORE - - - Response is generated from another node and is irrelevant for this node - Description: - This type is used to specify whether the frame processor is required to transmit the response part of the LIN frame. - Available via: - Lin↔_GeneralTypes.h -

External Assumption Req ID	External Assumption Text
SWS_Lin_00231	Name: - Lin_FrameDlType - Type: - uint8 - Range: - 1...8 - - Data length of a LIN Frame - Description: - This type is used to specify the number of SDU data bytes to copy. - Available via: - Lin_GeneralTypes.h -
SWS_Lin_00232	Name: - Lin_PduType - Type: - Structure - Element: - Lin_FramePidType - Pid - - Lin_FrameCsModelType - Cs - - Lin_FrameResponseType - Drc - - Lin_FrameDlType - Dl - - uint8* - SduPtr - - Description: - This Type is used to provide PID, checksum model, data length and SDU pointer from the LIN Interface to the LIN driver. - Available via: - Lin_GeneralTypes.h -
SWS_Lin_00233	Name: - Lin_StatusType - Type: - Enumeration - Range: - LIN_NOT_OK - - LIN frame operation return value. Development or production error occurred - LIN_TX_OK - - LIN frame operation return value. Successful transmission. - LIN_TX_BUSY - - LIN frame operation return value. Ongoing transmission (Header or Response). - LIN_TX_HEADER_ERROR - - LIN frame operation return value. Erroneous header transmission such as: - Mismatch between sent and read back data - Identifier parity error or - Physical bus error - LIN_TX_ERROR - - LIN frame operation return value. Erroneous response transmission such as: - Mismatch between sent and read back data - Physical bus error - LIN_RX_OK - - LIN frame operation return value. Reception of correct response. - LIN_RX_BUSY - - LIN frame operation return value. Ongoing reception: at least one response byte has been received, but the checksum byte has not been received. - LIN_RX_ERROR - - LIN frame operation return value. Erroneous response reception such as: - Framing error - Overrun error - Checksum error or - Short response - LIN_RX_NO_RESPONSE - - LIN frame operation return value. No response byte has been received so far. - LIN_OPERATIONAL - - LIN channel state return value. Normal operation; the related LIN channel is ready to transmit next header. No data from previous frame available (e.g. after initialization) - LIN_CH_SLEEP - - LIN channel state return value. Sleep state operation; in this state wake-up detection from slave nodes is enabled. - Description: - LIN operation states for a LIN channel or frame, as returned by the API service Lin_GetStatus(). - Available via: - Lin_GeneralTypes.h -
SWS_Lin_00106	The Lin module's environment shall not call any function of the Lin module before having called Lin_Init except Lin_GetVersionInfo.
SWS_Lin_00193	In case of receiving data the LIN Interface has to wait for the corresponding response part of the LIN frame by polling with the function Lin_GetStatus() after using the function Lin_SendFrame().
SWS_Lin_00194	The Lin module's environment shall only call Lin_SendFrame on a channel which is in state LIN_CH_OPERATIONAL or in one of the sub-states of LIN_CH_OPERATIONAL.
SWS_Lin_00239	In case of errors during header transmission, it is up to the implementer how to handle these errors (stop/continue transmission) and to decide if the corresponding response is valid or not.
SWS_Lin_00200	The return states LIN_TX_OK, LIN_TX_BUSY, LIN_TX_HEADER_ERROR, LIN_TX_ERROR, LIN_RX_OK, LIN_RX_BUSY, LIN_RX_ERROR, LIN_RX_NO_RESPONSE and LIN_OPERATIONAL are sub-states of the channel state LIN_CH_OPERATIONAL.
SWS_Lin_CONSTR_00279	DRAFT: LinChannel and LinTrcvChannel of one communication channel shall all reference the same ECUC partition.

External Assumption Req ID	External Assumption Text
SWS_Lin_91140	Name: - Lin_SlaveErrorType - Type: - Enumeration - Range: - LIN_ERR_HEADER - - - Error in header - LIN_ERR_RESP_STOPBIT - - - Framing error in response - LIN_ERR_RESP_CHKSUM - - - Checksum error - LIN_ERR_RESP_DATABIT - - - Monitoring error of transmitted data bit in response - LIN_ERR_NO_RESP - - - No response - LIN_ERR_INC_RESP - - - Incomplete response - Description: - This type represents the slave error types that are detected during header reception and response transmission / reception. - Available via: - Lin_GeneralTypes.h -
EA_RTD_00071	If interrupts are locked, a centralized function pair to lock and unlock interrupts shall be used.
EA_RTD_00082	When caches are enabled and data buffers are allocated in cacheable memory regions the buffers involved in DMA transfer shall be aligned with both start and end to cache line size. Note: Rationale: This ensures that no other buffers/variables compete for the same cache lines.
EA_RTD_00113	When RTD drivers are integrated with AutosarOS and User mode support is enabled, the integrator shall assure that the definition and declaration of all RTD functions needed to be called as trusted functions follow the naming convention Call<Function_Name>TRUSTED(parameter1,parameter2,...) in Integration/User code. They need to be visible in Os.h for the driver to call them. They will call RTD <Function_Name>() as trusted functions in OS specific manner.

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